

Navya reddy

AI ML

Day 5 task

1. Improve body land mark detection

The objective of this task was to improve human body landmark detection accuracy for virtual try-on and avatar generation applications.

Models Tested

The following pose estimation models were studied and tested:

MediaPipe

MoveNet

OpenPose

Testing Process

The models were tested using different types of images, including:

Full-body images

Side poses

Low-light conditions

Partial body occlusion

Multiple-person images

Observations

MediaPipe provided very fast real-time detection and worked efficiently for webcam-based applications. MoveNet delivered balanced performance between speed and accuracy. OpenPose produced highly accurate landmark detection, especially for complex poses and multiple people, but required higher computational resources.

2. Research Clothing Fitting Algorithms

Objective

The objective of this task was to study AI-based virtual clothing fitting systems and understand how garments are digitally aligned onto human bodies.

Algorithms Researched

The following virtual try-on algorithms and models were researched:

CP-VTON

VITON-HD

HR-VITON

TryOnDiffusion

Key Learnings

The study covered both 2D and 3D virtual try-on systems. 2D systems use image-based garment transfer and require less computational power, while 3D systems use body meshes and cloth simulation to generate more realistic outputs.

Important concepts studied included:

Human parsing

Cloth warping

Pose estimation

Texture preservation

Occlusion handling

Conclusion

Modern diffusion-based and GAN-based virtual try-on systems produce highly realistic clothing fitting results and improve texture quality significantly.

3. Test Virtual Try-On Datasets

Objective

The objective of this task was to analyze datasets used for training and testing virtual try-on systems.

Datasets Studied

The following datasets were researched:

DeepFashion

VITON Dataset

DressCode

Dataset Analysis

DeepFashion contains a large collection of clothing images with annotations and multiple clothing categories. The VITON dataset provides paired clothing and person images useful for virtual try-on tasks. DressCode offers modern clothing categories and high-quality fashion images.

Observations

DeepFashion was useful for large-scale fashion analysis. VITON was effective for virtual try-on experiments due to its paired structure. DressCode provided diverse and modern clothing styles suitable for advanced research.

Conclusion

DeepFashion and DressCode were identified as strong datasets for future virtual try-on model development because of their diversity and image quality.

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Run All Code

Draft Session (35m)

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DATASET TESTING REPORT
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1. FOLDER STRUCTURE:
input/
  notebooks/
    mohitgupta12/
      neetnetnet/
        hr-viton-try-on-clothes/
          __results__files/

2. FILE COUNT:
/kaggle/input/notebooks/neetnetnet/hr-viton-try-on-clothes: 4 files
/kaggle/input/notebooks/neetnetnet/hr-viton-try-on-clothes/__results__files: 1 files
TOTAL FILES: 5

3. CHECKING FOR CORRUPTED IMAGES:
Images Checked : 1
Corrupted Files: 0
Null/Bad Files : 0

4. IMAGE SIZE ANALYSIS:
Min Size : 743 x 658
Max Size : 743 x 658
Avg Size : 743 x 658

5. DATASET QUALITY:
Total images verified : 1
Corrupted : 0

```

Notebook

Input

+ Add Input Upload

NOTEBOOKS

HR-VITON - Try on clothes

Output (88KiB / 19.5GiB)

/kaggle/working

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Run All Code

Draft Session (35m)

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Corrupted : 0
Status: CLEAN - No corrupted files found

6. DISPLAYING SAMPLE IMAGES...

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Sample Images from Dataset

Notebook

Input

+ Add Input Upload

NOTEBOOKS

HR-VITON - Try on clothes

Output (88KiB / 19.5GiB)

/kaggle/working

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4.Create Simple Outfit Recommendation Logic

The objective of this task was to create a simple outfit recommendation system using rule-based logic.

Recommendation Logic

A basic recommendation system was created using clothing color and style matching rules.

Example:

Python

if shirt_color == "black":

recommend("white sneakers")

Features Considered

The recommendation system considered:

- Color combinations
- Clothing categories
- Casual and formal styles
- Seasonal preferences

Sample Recommendations

Examples of generated recommendations included:

Black T-shirt with blue jeans and white sneakers

White shirt with black trousers

Hoodie with joggers and sneakers

Conclusion

Simple rule-based recommendation systems can generate useful fashion suggestions and can later be improved using machine learning techniques.

5.Study AI Avatar Generation Workflows

The objective of this task was to study AI avatar generation systems and understand their workflow.

Workflow Studied

The avatar generation workflow included:

- 1.Image input collection
- 2.Face detection
- 3.Landmark extraction
- 4.Face embedding generation
- 5.AI avatar synthesis
- 6.Rendering and output generation

Technologies Researched

The following technologies and tools were studied:

StyleGAN

Stable Diffusion

MetaHuman Creator

Key Learnings:

GAN-based systems generate realistic human faces, while diffusion models improve image quality and detail generation. Accurate landmark extraction is important for creating realistic AI avatars.

Conclusion

AI avatar generation combines computer vision, deep learning, and image synthesis techniques to create realistic digital humans